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CS 76 Final Paper, Fall 2020

The Ethics of Recommender Systems

From streaming on Netflix to online shopping on Amazon, humans interact with recommender systems on a regular basis. In fact, a recent Mckinsey report found that 75% of Netflix streaming decisions and 35% of Amazon purchases come from recommendations as opposed to user search[1]. Recommender systems have become an integral part to many of the most commonly used digital services and applications. The prolific use of such algorithms, at first glance, appears benign; after all, suggesting movies or music to a user seems to only work to increase their utility. However, recommender systems pose serious ethical questions, specifically regarding data privacy and hazardous content. The nearly ubiquitous existence of these systems in our digital world gives rise to the question: Should recommender systems be optional?

It is first important to understand what recommender systems are and how they work. They are generally defined as systems that work to "propose ranked lists of items (that are subsets of a larger collection) according to their presumed relevance to individual users"[2], and this definition is most clearly demonstrated in the realm of e-commerce. The 'relevance' that predicates the system output can be determined from explicit or implicit user feedback, such as ratings of items, browsing patterns and commercial transactions. As a result, the user is suggested other items that follow their previous behavior with the system. For example, after watching a romance movie on Netflix, the user will be recommended similar movies in the romance genre; if the user chooses to rate the movie, these recommendations will be refined. However, it is not only the user's actions that facilitate the recommender systems. The systems

[1] https://www.mckinsey.com/industries/retail/our-insights/how-retailers-can-keep-up-with-consumers
[2] Jannach D., Zanker M., Ge M., Gröning M. (2012) Recommender Systems in Computer Science and Information Systems – A Landscape of Research. In: Huemer C., Lops P. (eds) E-Commerce and Web Technologies. EC-Web 2012. Lecture Notes in Business Information Processing, vol 123. Springer, Berlin, Heidelberg.

can also make use of the user's demographic data (relating to their age, race, sex etc.) and product attribute data.

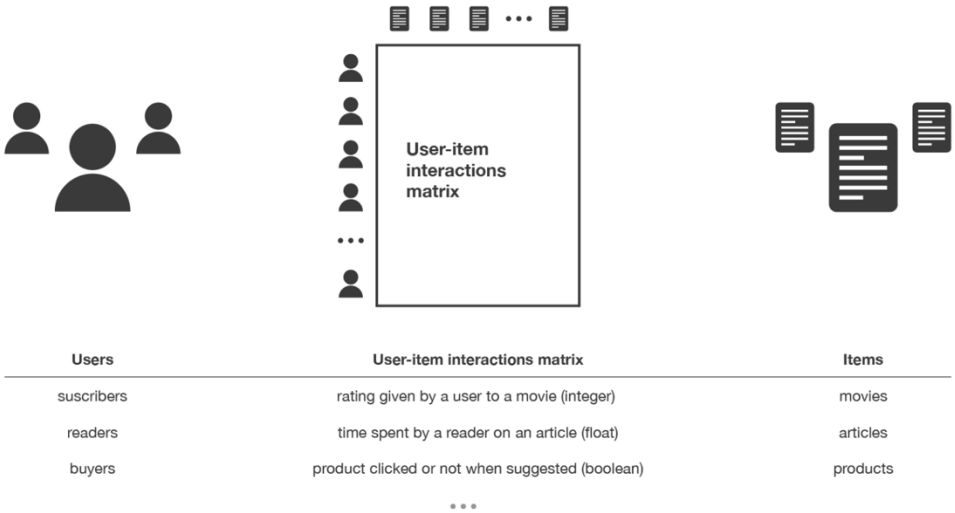

Figure 1: Illustration of the user-items interaction matrix[3]

The two main models of recommender systems are Collaborative filtering and Content based filtering. The Collaborative filtering approach is predicated on the assumption that a user will like the same items in the future that they liked in the past. User-item interactions are recorded and stored in a matrix. Within Collaborative filtering, a memory-based approach uses a 'nearest-neighbors' search: users and items are represented by past interactions and the system suggests the most popular items shared by the 'neighbors' of the user. A model based approach assumes an underlying model that explains the user-item interactions, and then uses this model to provide new suggestions. Collaborative filtering is advantageous because it does not require information specific to the user or item; it can provide accurate recommendations without fully "understanding" the nature of the item itself, and only requires keeping track of user-item interactions. On the other hand, Content Filtering uses item descriptions and user preferences to

[3] Rocca, Baptiste. "Introduction to Recommender Systems." *Medium*, Towards Data Science, 12 June 2019, towardsdatascience.com/introduction-to-recommender-systems-6c66cf15ada.

build a user profile. This method attempts to build a 'model' with these pieces of information that explain user decisions, such as how a user's gender may affect which movies they rate higher. In theory, the more the user interacts with this system, and the more information at hand, the more accurate the recommendations become. Many systems use a hybrid method of both the Collaborative and Content filtering techniques.[4]

Thus it is clear the stakeholders involved in recommendation systems are the providers, system, user, and society as a whole.[5] The provider is the company or service using the recommendation system, and typically wishes for their items to be recommended to users and to maintain the interest of the user. The system itself makes these recommendations, and wishes to function as efficiently and effectively as possible with its provided information. The user receives these recommendations and acts upon them; they wish to receive recommendations that will align with their taste and interests. Finally, society as a whole becomes a stakeholder;

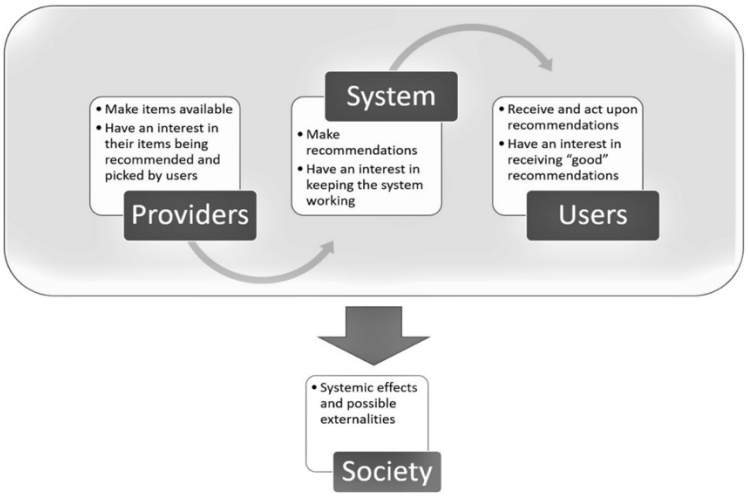

Figure 1: Stakeholders of Recommendation system (Milano, Silvia and Taddeo, Mariarosaria and Floridi, Luciano, Ethical Aspects of Multi-stakeholder Recommendation Systems *(November 25, 2019).)*

[4] Rocca, Baptiste. "Introduction to Recommender Systems." *Medium*, Towards Data Science, 12 June 2019, towardsdatascience.com/introduction-to-recommender-systems-6c66cf15ada.
[5] Milano, Silvia and Taddeo, Mariarosaria and Floridi, Luciano, *Ethical Aspects of Multi-stakeholder Recommendation Systems* (November 25, 2019).

recommendation systems may have systemic effects and possible ethical tradeoffs on a much broader scale than just the individual user. It follows that the ethical challenges of recommender systems are most clearly seen in impact on the stakeholders. More specifically, these systems may negatively impact the utility of stakeholders, or even violate their privacy.

A negative impact on the utility of stakeholders can come in many forms. An example with an immediate effect is when a recommendation is inaccurate, leading to dissatisfaction. However, there are far more nefarious consequences at hand. One of these is user addiction. An underlying purpose of recommendation systems is to pique and maintain user interest. Sean Parker, a founding member of Facebook, notes "The thought process that went into building these applications, Facebook being the first of them, … was all about: 'How do we consume as much of your time and conscious attention as possible.'"[6]. As a result, many users can find themselves spending hours and hours consuming digital media; such behavior has been popularized as "binge-watching" when referring to video streaming. Though this may seem harmless, researchers from the University of Texas, Austin found that binge-watching correlates with loneliness, depression and self-regulation deficiency. Additionally, researcher Yoon Hi Sung notes that "Physical fatigue and problems such as obesity and other health problems are related to binge-watching and they are a cause for concern. When binge-watching becomes rampant, viewers may start to neglect their work and their relationships with others"[7]. At first, this seems like a question of self-control. Can we really blame an algorithm for personal decisions? However, it is hard to ignore the role that recommendation systems play in this

[6] DeLeon, Haley. "The Ethical and Privacy Issues of Recommendation Engines on Media Platforms." *Medium*, Towards Data Science, 24 Apr. 2019, towardsdatascience.com/the-ethical-and-privacy-issues-of-recommendation-engines-on-media-platforms-9bea7bcb0abc.
[7] Sung, Y. et al. "A Bad Habit for Your Health? An Exploration of Psychological Factors for Binge Watching Behavior." (2015).

phenomenon; indeed, constantly supplying tailored content fuels this excessive media consumption. Stanford Professor and author Nir Eyal, who wrote the book "Hooked: How to Build Habit-Forming Products", is calling for ethical reform. He states, "There's nothing wrong with building products people want to use, but the power to design user behavior ought to come with a standard of ethical limitations"[8]. Though these systems help users stay hooked and entertained, it is clear that they come with the risk of negative health effects and the development of unhealthy behaviors. Furthermore, nudging users towards certain content and filtering this content can become, in extreme circumstances, an encroachment on individual autonomy. This intervention "can range from being benign (enabling individual agency and supporting better decision-making by filtering out irrelevant options), to being questionable (persuasion, nudging), and possibly manipulative"[9].

Furthermore, recommendation systems can lead to exposure to extreme content. As users are curated more and more content based on their previous interactions, the content becomes increasingly radical in order to keep users interested. This is specifically the case with YouTube, one of the most popular video-sharing platforms and a useful weapon for "political extremists"[10]. Social media platforms like YouTube " do not merely transmit content, but filter it on the basis of claiming to augment it, thereby making the content more relevant to its potential consumer", yet openly remain neutral regarding the nature of such content. In 2008 the YouTube Team

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[8] Eyal, Nir. "Want to Design User Behavior? Pass the 'Regret Test' First." *Nir and Far*, 23 Mar. 2020, www.nirandfar.com/regret-test/.

[9] Milano, Silvia and Taddeo, Mariarosaria and Floridi, Luciano, *Ethical Aspects of Multi-stakeholder Recommendation Systems* (November 25, 2019).

[10] O'Callaghan, D. et al. "Down the (White) Rabbit Hole: The Extreme Right and Online Recommender Systems." *Social Science Computer Review* 33 (2015): 459 - 478.

released the statement: "YouTube encourages free speech and defends everyone's right to express unpopular points of view…rather than stifle debate we allow our users to view all acceptable content and make up their own minds"[11]. YouTube's good intentions aside, an investigation conducted by The Wall Street Journal and Guillaume Chaslot, an ex-Google engineer, found that the sites recommendation system "fed far-right or far-left videos to users who watched relatively mainstream news sources," and was bias towards inflammatory content[12]. No matter where a user starts, as they continue to click and watch, their content became more extreme, driving them down a rabbit hole of volatile content, but YouTube and Google continue reap the financial benefits from advertisements. This is an especially precarious situation at present, given the growing number of adolescents and children who now have access to the internet, and YouTube being a fan-favorite among this demographic.

Finally, there is the ethical dilemma of user privacy. Many Recommendation systems use a Collaborative filtering or hybrid technique in which a 'model' is constructed for the user in order to personalize the recommendations. These models, however, often utilize personal information. This can range from the user's past purchasing or viewings to demographic details. This gives rise to certain privacy risks; once these data sets are stored to be used in the model, there is the chance they may be leaked. As a result, users may suffer a loss in utility, or worse, rights violations in which their private information is weaponized to threaten their autonomy and livelihood. In addition, privacy risks not only include the data collection itself, but also

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[11] MacKinnon, Rebecca. "WikiLeaks, Amazon and the New Threat to Internet Speech." *CNN*, Cable News Network, 3 Dec. 2010, www.cnn.com/2010/OPINION/12/02/mackinnon.wikileaks.amazon/index.html.

[12] Nicas, Jack. "How YouTube Drives People to the Internet's Darkest Corners." *The Wall Street Journal*, Dow Jones & Company, 7 Feb. 2018, www.wsj.com/articles/how-youtube-drives-viewers-to-the-internets-darkest-corners-1518020478.

recommendation outputs; an external agent who observes the recommendations given to a user can infer private information about the user. It is important to note that there have been "Privacy enhancing architectures" to help mitigate this, such as "storing user data in separate and decentralized databases, to minimize the risk of leaks"[13].

Social media, digital services and e-commerce are redefining how we interact and consume in our society. Recommendation systems have become integral to many digital media and e-commerce providers because they help keep the user engaged and satisfied . However, they also pose many ethical risks. It is therefore helpful to observe these risks from the lenses of a few major ethical theories: Deontology, Virtue Ethics and Utilitarianism. Deontology centers around moral law, and requires that these laws are applied "equally and impartially". Recommendation systems, however, actively perpetuate bias by targeting, filtering and tailoring outputs based on user 'models'. In fact, "it has been shown that Google shows ads for high-paying jobs to women much less frequently than to men"[14]. Though not every case of recommendations is malicious, the existence of bias within the functionality of a recommendation system implies that the 'law' is being applied unevenly amongst users. Virtue Ethics is focused on helping people achieve their end goals. On one hand, recommendation systems can help user's reach their goals faster by tailoring their content to their needs and taste. For example, a user who wishes to learn a musical instrument may search a video on YouTube for help. The recommended videos following this will facilitate and further their learning process. In contrast, the bias nature of recommendation systems can sometimes omit certain

[13] Milano, Silvia and Taddeo, Mariarosaria and Floridi, Luciano, *Ethical Aspects of Multi-stakeholder Recommendation Systems* (November 25, 2019).
[14] Burton, E., Goldsmith, J., Koenig, S., Kuipers, B., Mattei, N., & Walsh, T. (2017). Ethical Considerations in Artificial Intelligence Courses. *AI Magazine*, *38*(2), 22-34.

demographics from having access to the same content. For example, home loans and insurance companies can use recommendation systems "on a selective basis" to make them "unavailable to the residents of neighborhoods that are predominantly poor or are ethnic minorities"[15], thereby having the opposite effect of Virtue Ethics. Following this, Utilitarianism aims to bring the greatest utility to the greatest number if people possible. For the same reason that recommendation systems can be discriminatory, there will be cases in which users will be targeted (and likewise, not targeted), due to their demographics. As a result, otherwise talented and capable people may not be given the same access (or at the very least, ease of access) to content as others, and are thus denied the same opportunities to achieve and contribute to society. However, recommendation systems can also be extremely beneficial and helpful to large groups of people by making their digital experience easier and more efficient.

Overall, my personal opinion is that recommender systems should be optional; the user should be able to choose whether they want curated recommendations from a digital service or not. In doing so, the user would gain some personal autonomy over the recommendation system, avoid possible contact with harmful content and protect themselves against potential privacy violations. Firstly, allowing users to choose whether they want to receive recommended media or suggestions can help them avoid addictive and unhealthy behaviors. Though it is dependent on the user to make this choice themselves, having this option gives the user more autonomy over their experience when using the digital service. This further applies with regard to harmful or polarizing content; if users are allowed to choose whether a recommendation service is used or not, they gain a level of autonomy over the nature of the content they view. In both the case for

[15] Burton, E., Goldsmith, J., Koenig, S., Kuipers, B., Mattei, N., & Walsh, T. (2017). Ethical Considerations in Artificial Intelligence Courses. *AI Magazine*, *38*(2), 22-34.

addictive behaviors and radical content, an optional recommendation system would be a useful solution, especially when the users are young and susceptible. Much like parental controls, parents can turn off the recommendation system to help protect their children from overconsuming content or viewing harmful material. Finally, if users do not wish to run the risk of having their private information (such as demographics, streaming history or purchasing patterns) leaked, having the option to turn off a recommendation system can protect them from this because there will be no 'model' storing their information.

A possible counter argument to this position is that users should be responsible about how much they consume and the content they consume; one may argue that if a user develops addictive/dangerous behaviors or rhetoric it is their own fault for lacking the self-control and common-sense to avoid this fate. However, I would argue that this is merely shifts the responsibility on users alone; users are forced to carry this moral burden as they navigate the digital world and providers can simply wash their hands of any moral obligations. Though users should always be aware of their own actions, there are cases in which recommendation systems clearly violate user autonomy and can negatively impact their behavior, lifestyle and beliefs. Thus, optional recommendation systems put a piece of autonomy back in the hands of users, so that they can regain control over their digital experience.

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